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ARTICLE OF FOOTWEAR WITH DEVICE FOR EASE OF ENTRY

Abstract

A device for easing foot entry into an article of footwear includes a control bar that has a center segment, a medial side arm extending downwardly and forwardly from the center segment, and a lateral side arm spaced from the medial side arm and extending downwardly and forwardly from the center segment. The device includes a base underlying the control bar and connected to the medial side arm at a first joint and to the lateral side arm at a second joint. The device includes a medial anchor arm extending downwardly and forwardly from the first joint and partially under the foot-receiving void. The device includes a lateral anchor arm extending downwardly and forwardly from the second joint and partially under the foot-receiving void. The medial side arm and the lateral side arm resiliently bend to a loaded position when a downward force is applied to the center segment.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. Nonprovisional application Ser. No. 18/176,817, filed Mar. 1, 2023, which claims the benefit of priority to U.S. Provisional Application No. 63/346,612 filed May 27, 2022, and each of which are incorporated by reference in their entirety.

TECHNICAL FIELD

[0002] The present disclosure generally relates to a device for easing foot entry into an article of footwear and to an article of footwear that includes the device at a heel region of the article of footwear.

BACKGROUND

[0003] Traditionally, placing footwear on a foot often requires the use of one or both hands to enlarge the ankle opening of a footwear upper, and hold the rear portion during foot insertion, especially in the case of a relatively soft upper and/or an upper that does not have a heel counter secured to a flexible fabric rearward of the ankle opening. These steps may be difficult for some wearers, such as those with limited dexterity and young children and toddlers.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] The drawings described herein are for illustrative purposes only, are schematic in nature, and are intended to be exemplary rather than to limit the scope of the disclosure.

[0005] FIG. 1 is a lateral side view of an article of footwear having a device for easing foot entry shown in an unloaded, use position.

[0006] FIG. 2 is a medial side view of the article of footwear of FIG. 1.

[0007] FIG. 3 is a rear view of a portion of the article of footwear of FIG. 1, with the footwear upper not shown, and with a loaded, access position of the device shown in phantom.

[0008] FIG. 4 is a cross-sectional view of the article of footwear of FIG. 1 taken at lines 4-4 in FIG. 24 and showing a loaded position of the device in phantom.

[0009] FIG. 5 is a cross-sectional view of the article of footwear of FIG. 1 taken at lines 5-5 in FIG. 24.

[0010] FIG. 6 is a cross-sectional view of the article of footwear of FIG. 1 taken at lines 6-6 in FIG. 24.

[0011] FIG. 7 is a cross-sectional view of the article of footwear of FIG. 1 taken at lines 7-7 in FIG. 24.

[0012] FIG. 8 is a cross-sectional view of the article of footwear of FIG. 1 taken at lines 8-8 in FIG. 24.

[0013] FIG. 9 is a perspective view of the lateral side of the device included in the article of footwear of FIG. 1.

[0014] FIG. 10 is a perspective view of the medial side of the device included in the article of footwear of FIG. 1.

[0015] FIG. 11 is a top view of the device included in the article of footwear of FIG. 1.

[0016] FIG. **12** is a bottom view of the device included in the article of footwear of FIG. **1**.
[0017] FIG. **13** is a perspective view of the top of a bladder included in the article of footwear of FIG. **1**.
[0018] FIG. **14** is a perspective view of the bottom of the bladder included in the article of footwear of FIG. **1**.
[0019] FIG. **15** is a perspective view of the medial side of the device and bladder included in the article of footwear of FIG. **1**.
[0020] FIG. **16** is a perspective view of the lateral side of the device and bladder included in the article of footwear of FIG. **1**.
[0021] FIG. **17** is a perspective view of the bottom and medial side of a midsole included in the article of footwear of FIG. **1**.
[0022] FIG. **18** is a perspective view of the bottom and lateral side of the midsole included in the article of footwear of FIG. **1**.
[0023] FIG. **19** is a bottom view of the midsole included in the article of footwear of FIG. **1**.
[0024] FIG. **20** is a lateral side view of the midsole and the bladder included in the article of footwear of FIG. **1**.
[0025] FIG. **21** is a bottom view of the midsole and the device included in the article of footwear of FIG. **1**.
[0026] FIG. **22** is a lateral side view of the midsole and the device included in the article of footwear of FIG. **1**.
[0027] FIG. **23** is a bottom view of the midsole, the device, and the bladder included in the article of footwear of FIG. **1**.
[0028] FIG. **24** is a bottom view of the article of footwear of FIG. **1**.
[0029] FIG. **25** is a lateral side view of an alternative embodiment of an article of footwear having a device for easing foot entry shown in an unloaded, use position.
[0030] FIG. **26** is a medial side view of the article of footwear of FIG. **25**.
[0031] FIG. **27** is a rear view of a portion of the article of footwear of FIG. **25** and with a loaded, access position of the device shown in phantom.
[0032] FIG. **28** is a cross-sectional view of the article of footwear of FIG. **25** taken at lines **28-28** in FIG. **48** and showing a loaded position of the device in phantom.
[0033] FIG. **29** is a cross-sectional view of the article of footwear of FIG. **25** taken at lines **29-29** in FIG. **48**.
[0034] FIG. **30** is a cross-sectional view of the article of footwear of FIG. **25** taken at lines **30-30** in FIG. **48**.
[0035] FIG. **31** is a cross-sectional view of the article of footwear of FIG. **25** taken at lines **31-31** in FIG. **48**.
[0036] FIG. **32** is a cross-sectional view of the article of footwear of FIG. **25** taken at lines **32-32** in FIG. **48**.
[0037] FIG. **33** is a perspective view of the lateral side of the device included in the article of footwear of FIG. **25**.
[0038] FIG. **34** is a perspective view of the medial side of the device included in the article of footwear of FIG. **25**.
[0039] FIG. **35** is a top view of the device included in the article of footwear of FIG. **25**.
[0040] FIG. **36** is a bottom view of the device included in the article of footwear of FIG. **25**.
[0041] FIG. **37** is a perspective view of the top of a bladder included in the article of footwear of FIG. **25**.
[0042] FIG. **38** is a perspective view of the bottom of the bladder included in the article of footwear of FIG. **25**.
[0043] FIG. **39** is a perspective view of the medial side of the device and bladder included in the article of footwear of FIG. **25**.

[0044] FIG. **40** is a perspective view of the lateral side of the device and bladder included in the article of footwear of FIG. **25**.

[0045] FIG. **41** is a perspective view of the bottom and medial side of a midsole included in the article of footwear of FIG. **25**.

[0046] FIG. **42** is a perspective view of the bottom and lateral side of the midsole included in the article of footwear of FIG. **25**.

[0047] FIG. **43** is a bottom view of the midsole included in the article of footwear of FIG. **25**.

[0048] FIG. **44** is a lateral side view of the midsole and the bladder included in the article of footwear of FIG. **25**.

[0049] FIG. **45** is a bottom view of the midsole and the device included in the article of footwear of FIG. **25**.

[0050] FIG. **46** is a lateral side view of the midsole and the device included in the article of footwear of FIG. **25**.

[0051] FIG. **47** is a bottom view of the midsole, the device, and the bladder included in the article of footwear of FIG. **1**.

[0052] FIG. **48** is a bottom view of the article of footwear of FIG. **25**.

[0053] FIG. **49** is a lateral side view of an alternative embodiment of an article of footwear having a device for easing foot entry shown in an unloaded, use position.

[0054] FIG. **50** is a medial side view of the article of footwear of FIG. **49**.

[0055] FIG. **51** is a rear view of a portion of the article of footwear of FIG. **49** with a loaded, access position of the device shown in phantom.

[0056] FIG. **52** is a cross-sectional view of the article of footwear of FIG. **49** taken at lines **52-52** in FIG. **72** and showing a loaded position of the device in phantom.

[0057] FIG. **53** is a cross-sectional view of the article of footwear of FIG. **49** taken at lines **53-53** in FIG. **72**.

[0058] FIG. **54** is a cross-sectional view of the article of footwear of FIG. **49** taken at lines **54-54** in FIG. **72**.

[0059] FIG. **55** is a cross-sectional view of the article of footwear of FIG. **49** taken at lines **55-55** in FIG. **72**.

[0060] FIG. **56** is a cross-sectional view of the article of footwear of FIG. **49** taken at lines **56-56** in FIG. **72**.

[0061] FIG. **57** is a perspective view of the lateral side of the device included in the article of footwear of FIG. **49**.

[0062] FIG. **58** is a perspective view of the medial side of the device included in the article of footwear of FIG. **49**.

[0063] FIG. **59** is a top view of the device included in the article of footwear of FIG. **49**.

[0064] FIG. **60** is a bottom view of the device included in the article of footwear of FIG. **49**.

[0065] FIG. **61** is a perspective view of the top of a bladder included in the article of footwear of FIG. **49**.

[0066] FIG. **62** is a perspective view of the bottom of the bladder included in the article of footwear of FIG. **49**.

[0067] FIG. **63** is a perspective view of the medial side of the device and bladder included in the article of footwear of FIG. **49**.

[0068] FIG. **64** is a perspective view of the lateral side of the device and bladder included in the article of footwear of FIG. **49**.

[0069] FIG. **65** is a perspective view of the bottom and medial side of a midsole included in the article of footwear of FIG. **49**.

[0070] FIG. **66** is a perspective view of the bottom and lateral side of the midsole included in the article of footwear of FIG. **49**.

[0071] FIG. **67** is a bottom view of the midsole included in the article of footwear of FIG. **49**.

[0072] FIG. **68** is a lateral side view of the midsole and the bladder included in the article of footwear of FIG. **49**.

[0073] FIG. **69** is a bottom view of the midsole and the device included in the article of footwear of FIG. **49**.

[0074] FIG. **70** is a lateral side view of the midsole and the device included in the article of footwear of FIG. **49**.

[0075] FIG. **71** is a bottom view of the midsole, the device, and the bladder included in the article of footwear of FIG. **49**.

[0076] FIG. **72** is a bottom view of the article of footwear of FIG. **49**.

[0077] FIG. **73** is a perspective view of a medial side of the device of the article of footwear of FIG. **1** secured to a foam cushioning component.

[0078] FIG. **74** is a perspective view of a lateral side of the device and foam cushioning component of FIG. **73**.

[0079] FIG. **75** is a lateral side view of an alternative embodiment of an article of footwear having a device for easing foot entry shown in an unloaded, use position.

[0080] FIG. **76** is a bottom view of the article of footwear of FIG. **75**.

[0081] FIG. **77** is a cross-sectional view of the article of footwear of FIG. **75** taken at lines **77-77** in FIG. **76**.

[0082] FIG. **78** is a perspective view of an upper midsole layer included in the article of footwear of FIG. **75**.

[0083] FIG. **79** is a bottom view of a lower midsole layer included in the article of footwear of FIG. **75**.

[0084] FIG. **80** is a bottom view of a first outsole component included in the article of footwear of FIG. **75**.

[0085] FIG. **81** is a lateral side view of an alternative embodiment of an article of footwear having a device for easing foot entry shown in an unloaded, use position.

[0086] FIG. **82** is a bottom view of the article of footwear of FIG. **81**.

[0087] FIG. **83** is a cross-sectional view of the article of footwear of FIG. **81** taken at lines **83-83** in FIG. **82**.

[0088] FIG. **84** is a perspective view of an upper midsole layer included in the article of footwear of FIG. **81**.

[0089] FIG. **85** is a bottom view of a lower midsole layer included in the article of footwear of FIG. **81**.

[0090] FIG. **86** is a bottom view of a first outsole component included in the article of footwear of FIG. **81**.

[0091] FIG. **87** is a top view of a first outsole component of FIG. **86**.

[0092] FIG. **88** is a lateral side view of an alternative embodiment of an article of footwear having a device for easing foot entry shown in an unloaded, use position.

[0093] FIG. **89** is a bottom view of the article of footwear of FIG. **88**.

[0094] FIG. **90** is a cross-sectional view of the article of footwear of FIG. **88** taken at lines **90-90** in FIG. **89**.

[0095] FIG. **91** is a perspective view of a midsole included in the article of footwear of FIG. **88**.

[0096] FIG. **92** is a bottom view of a center outsole component included in the article of footwear of FIG. **88**.

[0097] FIG. **93** is a bottom view of a first outsole component included in the article of footwear of FIG. **88**.

DESCRIPTION

[0098] Various embodiments of a device for easing foot entry into an article of footwear are disclosed herein. The device may enable hands-free foot entry, such as by loading the device with the foot or otherwise to access a foot-receiving cavity from a rearward position, and sliding the foot

forward and downward into the foot-receiving cavity. Alternatively, a hand or hands may be used in conjunction with the device to ease foot entry. The footwear may be implemented for all sizes, and is especially useful for children and toddlers as it may enable them to independently don and doff their footwear. The various embodiments of articles of footwear disclosed herein are specifically configured for toddlers and children of preschool or grade school age, but may be scaled in size to adult footwear sizes.

[0099] In an example, a device for easing foot entry into an article of footwear and configured to surround a portion of a foot-receiving void at a heel region of an article of footwear. The device may include a control bar that has a center segment, a medial side arm extending downwardly and forwardly from the center segment, and a lateral side arm spaced from the medial side arm and extending downwardly and forwardly from the center segment. The device may include a base underlying the control bar and connected to the medial side arm at a first joint and to the lateral side arm at a second joint. The control bar and the base may be configured to surround a rear of the heel region with the medial side arm at a medial side of the article of footwear and the lateral side arm at a lateral side of the article of footwear. The device may include a medial anchor arm extending downwardly and forwardly from the first joint and partially under the foot-receiving void. The device may further include a lateral anchor arm extending downwardly and forwardly from the second joint and partially under the foot-receiving void. The medial side arm and the lateral side arm may resiliently bend to a loaded position when a downward force is applied to the center segment of the control bar, storing potential energy that returns the control bar to an unloaded position upon removal of the downward force.

[0100] An article of footwear that includes the device may include an upper and a sole structure that is secured to and underlies the upper. The upper and the sole structure may define the foot-receiving void. The control bar and the base may be configured to surround a rear of the upper with the medial side arm at a medial side of the upper, the lateral side arm at a lateral side of the upper and the base secured to an upper surface of the sole structure. The medial anchor arm may extend downwardly and forwardly from the first joint and partially under the sole structure and the foot-receiving void. The lateral anchor arm may extend downwardly and forwardly from the second joint and partially under the sole structure and the foot-receiving void. The medial side arm and the lateral side arm may resiliently bend to a loaded position when a downward force is applied to the center segment of the control bar, storing potential energy that returns the control bar to an unloaded position upon removal of the downward force. The rear of the upper moves with the center segment of the control bar. Accordingly, the foot-receiving void opens to a greater extent from the rear direction when the control bar is depressed, enabling easier foot entry.

[0101] In some implementations, the sole structure to which the device is fixed may be one or more foam midsole layers. Alternatively, in one or more implementations, the sole structure may include a bladder defining an interior cavity and configured to retain a fluid in the interior cavity. The base may be secured to an upper surface of the bladder.

[0102] The device may include one or more features that protect the bladder from external forces, such as may occur during wear. For example, in an aspect, the base may include a medial side projection extending transversely outwardly and downwardly over the bladder from a medial side of the base rearward of the first joint. The medial side projection is not fixed to the bladder. In another aspect, the base may include a lateral side projection extending transversely outwardly and downwardly over the bladder from a lateral side of the base rearward of the first joint. The lateral side projection is not fixed to the bladder. In still another aspect, the base may include a rear projection extending rearwardly and downwardly over the bladder from a rear of the base. The rear projection is not fixed to the bladder. The medial side projection, the lateral side projection, and the rear projection, in embodiments that include one or all of these, enable the bladder to resiliently deform outwardly under loading without increased stress on the bladder that would otherwise be caused if the projection(s) were fixed to the bladder.

[0103] In another aspect, an exterior rear surface of the base above the rear projection defines a concavity. This concavity may enable a wearer to insert the toe area of their opposite foot (or footwear on the opposite foot) into the concavity to grip the base and press downward on the base while pulling their foot out of the article of footwear. It is preferable to place the foot on the base than on the bladder, and the concavity encourages this placement. If the rear projection projecting downwardly and outwardly from the rear of the base is provided, it may be directly under the concavity, further protecting the bladder when the article of footwear is removed in this manner.

[0104] The sole structure may also include a midsole extending forward of the base. The medial anchor arm may nest in a medial recess in an exterior surface of the midsole and may be outwardly exposed. The lateral anchor arm may nest in a lateral recess in the exterior surface of the midsole and may be outwardly exposed. In some implementations, the midsole may extend rearwardly and transversely inwardly of the base to a rear portion of the base and may be secured to the upper surface of the bladder.

[0105] In one or more implementations, a rear surface of the medial anchor arm may define a downwardly extending rear wall and a front surface of the bladder may nest against the rear wall of the medial anchor arm.

[0106] Similarly, in one or more implementations, a rear surface of the lateral anchor arm may define a downwardly extending rear wall and a front surface of the bladder may nest against the rear wall of the lateral anchor arm.

[0107] In one or more implementations, a portion of the medial anchor arm and a portion of the lateral anchor arm may be exposed at a ground-facing surface of the sole structure.

[0108] The device may thus wrap over, around, and forward of the bladder and may be secured to both the upper and the midsole to fully integrate the device into the footwear.

[0109] The above features and advantages and other features and advantages of the present teachings are readily apparent from the following detailed description of the modes for carrying out the present teachings when taken in connection with the accompanying drawings. It should be understood that even though in the following the embodiments may be separately described, single features thereof may be combined in additional embodiments.

[0110] Referring to the drawings, wherein like reference numbers refer to like components, FIG. 1 shows an article of footwear **10** that includes a device **12** for easing foot entry into a foot-receiving cavity **14** of the article of footwear **10**. The article of footwear **10** includes a footwear upper **16** and a sole structure **18** underlying the footwear upper **16**. The footwear upper **16** and the sole structure **18** together define the foot-receiving cavity **14**, and the footwear upper **16** defines an ankle opening **20** through which a foot is inserted into the foot-receiving cavity **14**. The footwear upper **16** and the sole structure **18** are described in greater detail herein.

[0111] The article of footwear **10** is depicted as an athletic shoe and may be for a child, such as a child of grade school age, but within the scope of the present teachings the article of footwear **10** may be sized for an adult, and the features of the article of footwear **10** may be implemented in a dress shoe, a work shoe, a sandal, a slipper, a boot, or any other category of footwear. The article of footwear **10** shown is for a right foot. A pair of footwear includes the article of footwear **10**, and an article of footwear that is a mirror image of the article of footwear **10** for a left foot.

[0112] The article of footwear **10** includes a heel region **22**, a midfoot region **24**, and a forefoot region **26**. The heel region **22** generally includes portions of the article of footwear **10** corresponding with rear portions of a human foot, including the calcaneus bone, when the human foot is supported on the sole structure **18** in the foot-receiving cavity **14** and is a size corresponding with the article of footwear **10**. The forefoot region **26** of the article of footwear **10** generally includes portions of the article of footwear **10** corresponding with the toes and the joints connecting the metatarsals with the phalanges of the human foot (interchangeably referred to herein as the “metatarsal-phalangeal joints” or “MPJ” joints). The midfoot region **24** of the article of footwear **10** is disposed between the heel region **22** and the forefoot region **26** and generally includes

portions of the article of footwear **10** corresponding with an arch area of the human foot, including the navicular joint.

[0113] The device **12** is configured to surround a portion of the foot-receiving cavity **14** at the heel region **22** of the article of footwear **10**. The device **12** is a single, unitary, one-piece component that includes a control bar **30**, a base **32**, a medial anchor arm **34** (see FIG. 2), and a lateral anchor arm **36**. For example, the device **12** may be injection molded as a single, unitary, one-piece component. The material of the device **12** is selected to provide the ability to elastically deform by elastic bending as described herein, and store potential energy, such as elastic energy, that returns the device **12** to the unloaded position shown in FIGS. 1 and 2. Example materials include plastics (such as thermoplastics), composites, and nylon. Another example material is a polyether block amide such as PEBAX® available from Arkema, Inc. in King of Prussia, Pennsylvania USA.

[0114] The control bar **30** has a center segment **30A**, a medial side arm **30B** extending downwardly and forwardly from the center segment **30A** (see FIG. 2), and a lateral side arm **30C** spaced from the medial side arm **30B** and also extending downwardly and forwardly from the center segment **30A**. The center segment **30A** has a thinned portion **31**. Stated differently, the portion of the center segment **30A** that surrounds the thinned portion is thicker from an inner side to an outer (exterior) side than at the thinned portion **31**, as best shown in FIG. 4, and the thinned portion **31** is thus relatively thin. The thinned portion **31** may be created when the device **12** is molded or otherwise formed. The thinned portion **31** is sufficiently thin that the upper **16** is stitched to the device **12** by stitching **33** that extends through the upper **16** and through the thinned portion **31**. This enables the rear portion of the upper **16** to move with the center segment **30A** of the control bar **30** to the access position as described herein.

[0115] The base **32** underlies the control bar **30** when the article of footwear **10** is disposed with the sole structure **18** resting on a level surface and the upper **16** is above the sole structure **18**, as in a position when a wearer is standing. The base **32** is connected to the medial side arm **30B** at a first joint **35** (see FIG. 2) and to the lateral side arm **30C** at a second joint **37** (see FIG. 1). The base **32** is continuous and extends between and connects to the medial side arm **30B** and the lateral side arm **30C**. The base **32** is continuous, in that it is without breaks or connections through other components in extending from the medial side arm **30B** and the lateral side arm **30C**. More specifically, the base **32** has a center segment **32A**, a medial base arm **32B**, and a lateral base arm **32C**. The medial base arm **32B** is spaced apart from the lateral base arm **32C** and both extend from the center segment **32A**. The first joint **35** is the connection of the medial base arm **32B** to the medial side arm **30B**. The second joint **37** is the connection of the lateral base arm **32C** to the lateral side arm **30C**.

[0116] The control bar **12** has an arced shape from the first joint **35** to the second joint **37**. Similarly, the base **32** has an arced shape from the first joint **35** to the second joint **37**. The device defines a gap **38** (indicated in FIGS. 9 and 10) between the control bar **30** and the base **32** extending around the rear of the upper **16** from the first joint **35** to the second joint **37**. With this arrangement, the control bar **30** and the base **32** are configured as a full elliptical leaf spring. The article of footwear **10** is characterized by an absence of a heel counter. Stated differently, no heel counter is included at the gap **38**. Instead, a portion of the upper **16** is only a single layer **16A** of a flexible material between the center segment **30A** of the control bar **30** and a center segment **32A** of the base **32**. One or more additional inner or outer layers may be secured to the single layer **16A** with stitching **40** nearer to the control bar **30** to provide greater cushion between the control bar **30** and the foot and to surround a cushioned collar component **42** above the center segment **30A** of the control bar **30** and above a ledge **44** of the device **12** disposed at an upper extent of the center segment **30A**, as shown in FIG. 4. In an example, the cushioned collar component **42** may be foam. The device **12** functions at least in some respects as a heel counter in that it helps to retain a wearer's heel in position atop the heel region **22** of the sole structure **18**, limiting or preventing medial or lateral displacement during use.

[0117] The control bar **30** and the base **32** are configured to surround a rear **65** of the heel region **22** with the medial side arm **30B** at a medial side **48** of the article of footwear **10** and the lateral side arm **30C** at a lateral side **50** of the article of footwear **10**. More specifically, the medial side arm **30B** is outward of a medial side **48** of the upper **16** and the lateral side arm **30C** is outward of a lateral side **50** of the upper **16**. The side arms **30B**, **30C** are shown exposed, outward of an outermost exterior of the upper **16**. The medial side arm **30B** is not secured to the medial side **48** of the upper **16** and the lateral side arm **30C** is not secured to the lateral side **50** of the upper **16**. This allows the side arms **30B**, **30C** to bow outwardly under loading during foot entry, as shown in phantom in FIG. 3, without being restricted by the sides of the upper **16**.

[0118] The medial anchor arm **34** extends downwardly and forwardly from the first joint **35** and partially under the sole structure **18** and the foot-receiving cavity **14** as best shown in FIGS. 2, 7, 8, 23 and 24. The lateral anchor arm **36** extends downwardly and forwardly from the second joint **37** and partially under the sole structure **18** and the foot-receiving cavity **14** as best shown in FIGS. 1, 7, 8, 23 and 24.

[0119] The sole structure **18** includes a bladder **52** that defines an interior cavity **55** and is configured to retain fluid in the interior cavity **55**. The sole structure **18** further includes a midsole **54**, a first outsole component **56**, and a second outsole component **58** each of which is discussed in further detail herein. The base **32** is disposed on the bladder **52** and is secured to an upper surface **57** of the bladder **52** as shown, for example in FIG. 4. The medial anchor arm **34** and the lateral anchor arm **36** extend forward of the bladder **52** and are secured to the midsole **54** immediately forward of the bladder **52** as discussed herein.

[0120] The control bar **30** is biased to an unloaded position shown in FIGS. 1-4. The unloaded position is also referred to herein as an unstressed position. The control bar **30** is internally biased to the unloaded position by its material in its formed state. Stated differently, the material of the control bar **30** is sufficiently rigid that it remains in the unloaded position in its formed state without external loads applied to it, and will return to the unloaded position after elastically bending under an applied force due to its resiliency.

[0121] The medial side arm **30B** and the lateral side arm **30C** resiliently bend to a loaded position when a downward force **F** (shown in FIGS. 3 and 4) is applied to the center segment **30A** of the control bar **30**, storing potential energy that returns the control bar **30** to the unloaded position upon removal of the downward force **F**.

[0122] In the unloaded position, the center segment **30A** is a first distance **D1** from the base **32**, as indicated in FIG. 3 by a distance **D1** from the top of the center segment **30A** to the bottom of the center segment **32A** of the base **32**. The unloaded position is the position of the device **12** in a relaxed, unloaded state (i.e., without the downward force **F** applied to the control bar **30**).

[0123] The control bar **30** can be depressed under the applied force **F**, representing the force applied by a foot **62** during insertion of the foot **62** into a foot-receiving cavity **14** (see FIGS. 3 and 4) of the article of footwear **10**. As shown in FIG. 4, contact of the foot **62** may be with the upper **16** above the cushioned collar component **42** that extends above the center segment **30A**, and the force **F** may be transmitted to the control bar **30** through the upper **16** and cushioned collar component **42**. The ledge **44** extends in a fore-aft (longitudinal) direction rearward from the center segment **30A** to provide a greater area over which the force **F** is dispersed and, likewise, a greater surface area on the foot **62** over which reaction forces are dispersed. The ledge **44** prevents potential foot discomfort that may otherwise occur if the upper extent of the center segment **30A** had less surface area, as this would concentrate reaction forces on the foot **62**.

[0124] The force **F** is first applied by the toes or ball of the foot **62**, and then the foot **62** slides forward into the foot-receiving cavity **14** as shown in FIG. 4 as the control bar **30** depresses to the loaded position. The force **F** could instead be applied by a hand or hands. The device **12** is indicated with phantom lines and reference number **12A** in FIGS. 3 and 4 when in the depressed, loaded position. When loaded in this manner, the control bar **30** elastically bends to the loaded

position in which the center segment **30A** is a second distance **D2** from the base **32**, as shown in FIG. **3**. The second distance **D2** is less than the first distance **D1** and is measured from the top of the control bar **30** at the center segment **30A** to the bottom of the base **32** at the center segment **32A**. The difference between the distances **D1**, **D2**, is the deflection of the device **12**, which is dependent upon the level of force **F** applied. The device **12** is configured so that when it is depressed under the force **F** to the loaded position **D2**, the side arms **30B**, **30C** elastically bend near the joints **35**, **37**, storing elastic energy. When the force **F** is removed, such as when the foot **62** is fully inserted, the stored elastic energy returns the control bar **30** to the unloaded position. In FIG. **3**, only the device **12** and the sole structure **18** are shown. The upper **16** is removed for clarity in showing the positions of the device **12**, **12A**.

[0125] Traditionally, slipping a foot into a foot-receiving cavity of an upper often requires the use of one or both hands to stretch the ankle opening **20** and hold the rear portion of the upper during foot insertion, especially in the case of a relatively soft upper and/or an upper that does not have a heel counter secured to the flexible fabric rearward of the ankle opening. The device **12** alleviates these issues, and allows the foot **62** to enter into the foot-receiving cavity **14** without the use of hands or other tools. Entry may be gained using only the foot **62**. Children and toddlers may learn to operate the device **12** in this manner to don the article of footwear **10**, or may use their hand or hands, or an adult may use their hand or hands, for example, to assist the child or toddler with donning the article of footwear **10**.

[0126] The footwear **10** is shown in FIGS. **1** and **2** with a lace **64** that extends through lace-receiving features such as apertures **66** and anchoring loops **68**. The lace **64** may be tightened to adjust the girth of the foot-receiving cavity **14** over a tongue portion **16B** of the upper **16**. Other fastening features instead of or in addition to a lace may be used, or the upper **16** may be a sock-like upper with sufficient elasticity to be used without a lace or fastening features. The lace **64** is not shown in FIG. **4** for clarity.

[0127] As shown in FIG. **4**, the upper **16** is secured to the center segment **30A** with the stitching **33**, and moves down with the control bar **30** to the position **16AA** shown in phantom in FIG. **3**. The ankle opening **20** is indicated as **20A** when the device **12** is in the loaded position. The flexibility of the single layer **16A** helps to allow the upper **16** to temporarily fold or deform. In FIG. **3** the single layer **16A** is shown folded inward at the position **16AA**, but could instead fold outward, or may fold at more than one location, so that the upper **16** folds partially inward and partially outward. The stored elastic energy due to the bias of the device **12** automatically returns the device **12** to the unloaded position when the foot **62** moves fully into the foot-receiving cavity **14**, causing the upper **16** to be automatically pulled up over the back of the foot **62**.

[0128] Additionally, the relative dimensions and shape of the device **12** at the joints **35**, **37** and at the side arms **30B**, **30C** contributes to the spring-biased nature of the device **12**, and its ability to elastically deform under a desired amount of loading and return to its original unloaded position. For example, the side arms **30B**, **30C** each have a thickness greater than a width at the respective joint **35**, **37**, where the thickness is measured in the fore-aft (longitudinal) direction of the footwear **10** and the width is measured in the medial-lateral (transverse) direction of the footwear **10**. The joints **35**, **37** are much thicker than the side arms **30B**, **30C** so that bending will occur at the side arms **30B**, **30C** under sufficient loading. The thickness and the length of the side arms **30B**, **30C** are selected along with the material of the device to result in a desired level of force **F** required to resiliently bend the device **12** to the loaded position.

[0129] As shown in FIG. **3**, the device **12** is also configured to widen as it is moved from the unloaded position to the loaded position. This helps ease insertion of the foot **62** into the foot-receiving cavity **14**, as the side arms **30B**, **30C** bow apart from one another to the positions **30BB** and **30CC** when the control bar **30** is depressed, allowing the upper **16** to be stretched outward by the entering foot **62** without interference from the side arms **30B**, **30C**.

[0130] As best shown in FIGS. **1-3** and **5**, the midsole **54** extends forward of the base **32** and

forward of the entire device **12**. As shown in FIGS. **4**, **7**, and **8**, the midsole **54** extends rearwardly and transversely inwardly of the base **32** to a rear portion of the base **32** (e.g., to the inner side of the center segment **32A** of the base **32**) and is secured to the upper surface **57** of the bladder **52** forward of the center segment **32A** of the base **32**. The midsole **54** may be molded with contours that specifically fit to the contours of the bladder **52**. For example, as shown in FIG. **13**, the bladder **52** has an inflation port **61**, and the lower surface of the midsole **54** has a recess **63** shaped to receive the inflation port **61**.

[0131] FIGS. **6** and **24** show that a portion of the medial anchor arm **34** and a portion of the lateral anchor arm **36** are exposed at a ground-facing surface **70** of the sole structure **18**. The outsole components **56**, **58** and the portions of the medial anchor arm **34** and lateral anchor arm **36** thus establish the ground-facing surface **70**. The bladder **52** is covered at the ground-facing surface **70** by the first outsole component **56** and the second outsole component **58** so that it is protected. As shown in FIGS. **14** and **23**, the bottom surface **74** of the inflated portion of the bladder **52** is arcuate, extending around a periphery of the heel region **22**. The first outsole component **56** is also arcuate and is secured to and covers the bottom surface **74**, extending up to the flange **82** shown in FIG. **4**. The first outsole component **56** is also secured to a bottom surface of the medial anchor arm **34** and the lateral anchor arm **36** forward of the bladder **52**, as is evident by comparing FIG. **23** (showing the bladder **52**, the device **12**, and the midsole **54** without the outsole components **56**, **58**) to FIG. **24**, which shows the outsole components **56**, **58**. When the first outsole component **56** is secured to the bladder **52** and to the medial anchor arm **34** and lateral anchor arm **36**, a forwardmost portion of the medial anchor arm **34** and a forwardmost portion of the lateral anchor arm **36** are not covered by the first outsole component **56** and are exposed at the ground-facing surface of the sole structure **18**.

[0132] The second outsole component **58** is secured to the bladder **52** inward of the first outsole component **56**. For example, as best shown in FIG. **4**, the second outsole component **58** is secured to a bottom surface **76** of an uninflated portion of the bladder **52** where a first polymeric sheet **77**, that may be referred to as an upper sheet, and a second polymeric sheet **78**, that may be referred to as a lower sheet, are secured to one another. This uninflated portion is referred to as webbing **84**, as discussed herein. The second outsole component **58** also extends forward under the midsole **54** and is secured to a bottom surface **80** of the midsole **54** forward of the bladder **52**.

[0133] The bladder **52** may be thermoformed from the first polymeric sheet **77** and the second polymeric sheet **78**. Alternatively, the bladder **52** could be blow-molded from a pre-form polymeric material. The bladder **52** can be formed from any of various polymeric materials that retain a fluid at a predetermined pressure, including a fluid that is a gas, such as air, nitrogen, or another gas. As used herein, a “fluid” includes a gas, including air, an inert gas such as nitrogen, or another gas. Accordingly, “fluid-filled” includes “gas-filled”. For example, the bladder **52** can be a TPU material, a urethane, polyurethane, polyester, polyester polyurethane, and/or polyether polyurethane. Moreover, the sheets **77**, **78** may be laminate membranes formed from thin films having one or more first layers that comprise thermoplastic polyurethane layers and that alternate with one or more second layers, also referred to herein as barrier layers, gas barrier polymers, or gas barrier layers. In another example, the layers may include ethylene-vinyl alcohol copolymer, thermoplastic polyurethane, and a regrind material of the ethylene-vinyl alcohol copolymer and thermoplastic polyurethane. In the embodiment shown, the sheets **77**, **78** are transparent.

[0134] The sheets **77**, **78** are bonded to one another at a periphery of the bladder **52**, such as at an outer flange **82**. The sheets **77**, **78** are also bonded to one another at various intermediate bond locations, referred to as webbing **84**. The bonded sheets **77**, **78** form the fluid-filled interior cavity **55** which is fluid-tight, and may be pressurized or unpressurized.

[0135] In addition to the outsole components **56**, **58** covering the bottom surface of the bladder **52** to protect the bladder **52**, the device **12** also has many features that interfit with the bladder **52**, and may serve to protect the bladder **52** while enabling the bladder to perform its cushioning function.

For example, the upper surface **57** of the inflated portion of the bladder **52** is a convex upper surface, as shown in FIGS. **4**, **7**, **8**, and **13**. A lower surface **86** of the base **32** (shown in FIGS. **4**, **7**, **8**, and **12**) is concave where it interfits with and is secured to the convex upper surface **57** of the bladder **52**.

[0136] As shown in FIGS. **4**, **8**, and **12**, the lower surface **86** of the base **32** has an inner edge **88** and an outer edge **90** and is concave in a transverse direction of the article of footwear **10** and in a longitudinal direction of the article of footwear **10** from the inner edge **88** to the outer edge **90**, with the outer edge **90** lower than the inner edge **88**. The geometrically-mating concave lower surface **86** and convex upper surface **57** enables the lower surface **86** to securely bond to the upper surface **57** of the bladder **52**. Additionally, because the outer edge **90** extends relatively low (e.g., lower than the inner edge **88**) it is able to protect and shield more of the outer surface of the bladder **52** at the sides and rear of the article of footwear **10**.

[0137] The base **32** also has one or more projections that further shield the bladder **52** without unduly restricting the tendency for the bladder **52** to resiliently deform laterally outward and longitudinally in the fore aft direction as it compresses under loading. For example, the base **32** has a medial side projection **92** extending transversely outwardly and downwardly over the bladder **52** from a medial side of the base **32** rearward of the first joint **35**, as shown in FIGS. **2**, **7**, **8**, and **15**. The medial side projection **92** extends from the medial base arm **32B**. The medial side projection **92** is not fixed to the bladder **52**, as is best indicated in FIG. **8** by the small gap between the lower surface **86** and the upper surface **57** towards the outer edge **90**. Stated differently, the medial side projection **92** is not bonded to the bladder **52**.

[0138] Similarly, the base **32** has a lateral side projection **94** extending transversely outwardly and downwardly over the bladder **52** from a lateral side of the base **32** rearward of the second joint **37**, as shown in FIGS. **1**, **7**, **8**, and **9**, for example. The lateral side projection **94** extends from the lateral base arm **32C**. The lateral side projection **94** is not fixed to the bladder **52**, as is best indicated in FIG. **8** by the small gap between the lower surface **86** and the upper surface **57** towards the outer edge **90**. Stated differently, the lateral side projection **94** is not bonded to the bladder **52**.

[0139] The base **32** also has a rear projection **96** extending rearwardly and downwardly over the bladder **52** from a rear of the base **32** as best shown in FIGS. **1**, **3**, **4**, and **15**, for example. The rear projection **96** extends from the center segment **32A** of the base **32**. In some embodiments, the rear projection **96** is not fixed to the bladder **52**. Stated differently, the rear projection **96** is not bonded to the bladder **52**.

[0140] The downward and outward orientation of the projections **92**, **94**, and **96** function to shield portions of the bladder **52** directly inward of the projections **92**, **94**, and **96** from external forces and objects that may be encountered during use, such as when a child wearing the footwear **10** is playing, climbing, etc. Additionally, any of the projections **92**, **94**, and **96** that are not bonded to the bladder **52** enable the bladder **52** to resiliently deform under compression during wear without restraining movement of the bladder **52** along the inner surface of the projection, avoiding stresses on the bladder **52** that could occur if the bladder **52** were bonded or otherwise fixed to the inner surface of the projection.

[0141] An additional feature of the device **12** that lends utility while protecting the bladder **52** is a concavity **98** defined by the base **32** at an exterior rear surface **100** of the base **32** above the rear projection **96**. The concavity **98** is indicated in FIGS. **2-4**, **9**, and **10**, for example. The concavity **98** eases removal of the article of footwear **10** by providing the wearer with a convenient location to press downward and forward on the base **32** while attempting to withdraw the foot **62** from the foot-receiving cavity **14**. The concave configuration provides better leverage as a portion of the footwear on the opposite foot (or the opposite foot itself), such as the tip of the forefoot, may be inserted into the concavity **98**. By locating the rear projection **96** directly below the concavity **98**, the bladder **52** is better protected from contact with the foot in the concavity **98**.

[0142] The article of footwear **10** is also configured so that the bladder **52**, the medial and lateral

anchor arms **34**, **36** of the device **12**, and the midsole **54** cooperatively fit together. For example, referring to FIG. **17**, the midsole **54** includes a medial recess **102** in an exterior surface **104** of the midsole **54**. FIG. **18** shows that the midsole **54** includes a lateral recess **106** in the exterior surface **104** of the midsole **54**. For recesses **102**, **106** are at the outer sides of the midsole **54** and extend under the midsole **54**. The device **12** is configured to fit to the midsole **54** within the recesses **102**, **106**. For example, referring to FIG. **11**, the forward upper surface **111** of the medial anchor arm **34** protrudes inward and downward, matching the contour of the medial recess **102** so that the medial anchor arm **34** nests in the medial recess **102**. The medial anchor arm **34** may be bonded to the midsole **54** at the medial recess **102**. Similarly, the forward upper surface **113** of the lateral anchor arm **36** protrudes inward and downward, matching the contour of the lateral recess **106** so that the lateral anchor arm **36** nests in the lateral recess **106**. The lateral anchor arm **36** may be bonded to the midsole **54** at the medial recess **102**. FIG. **21** illustrates the anchor arms **34**, **36** nesting in the respective recesses **102**, **106**. FIG. **22** best illustrates the lateral anchor arm **36** nesting in the lateral recess **106**.

[0143] FIGS. **1**, **2**, **21**, and **22** also show that the medial anchor arm **34** and the lateral anchor arm **36** are outwardly exposed. For example, the medial anchor arm **34** and the lateral anchor arm **36** are outwardly exposed at the medial side **48** and the lateral side **50** of the article of footwear **10**, respectively. As previously discussed, the medial anchor arm **34** and the lateral anchor arm **36** are also outwardly exposed at the bottom of the article of footwear **10**, with a portion of each still being exposed even after the first outsole component **56** is secured to the bladder **52** and to a portion of the medial anchor arm **34** and a portion of the lateral anchor arm **36**.

[0144] In addition to the nesting of the device **12** in the recesses **102**, **106** at the exterior of the midsole **54**, the device **12** is also configured so that the bladder **52** nests against the medial anchor arm **34** and the lateral anchor arm **36**. For example, referring to FIGS. **10** and **12**, a rear surface **114** of the medial anchor arm **34** is concave and defines a downwardly extending rear wall **116**. Similarly, a rear surface **118** of the lateral anchor arm **36** is concave and defines a downwardly extending rear wall **120**. The walls **116**, **120** are also indicated in FIGS. **21** and **22**.

[0145] As best shown in FIG. **13**, the bladder **52** has a convex front surface **122** near the medial side **48** of the bladder **52** and a convex front surface **124** near the lateral side **50** of the bladder **52**. FIG. **15** shows that the convex front surface **122** of the bladder **52** nests against the rear wall **116** of the medial base arm **32B**. The front surface **122** may be bonded to the rear wall **116** or may simply confront the rear wall **116**. FIG. **16** shows that the convex front surface **124** of the bladder **52** nests against the rear wall **120**. The front surface **124** may be bonded to the rear wall **120** or may simply confront it. The downward and forward extension of the medial anchor arm **34** and the lateral anchor arm **36** from the base **32** as well as the inward extension of the arms **34**, **36** enable the medial anchor arm **34** and the lateral anchor arm **36** to be sandwiched between the bladder **52** and the midsole **54** at the front of the bladder **52**.

[0146] Accordingly, the device **12** is thus secured to the upper **16**, to the bladder **52**, and to the midsole **54** and wraps around the upper **16** from the rear of the upper **16**, along the medial and lateral sides **48**, **50**, and extends forward of the bladder **52**.

[0147] FIG. **25** is a lateral side view of an alternative embodiment of an article of footwear **110** having a device **112** for easing foot entry into the foot-receiving cavity **14** of the article of footwear **110**. The device **112** is shown in an unloaded, use position. The article of footwear **110** and the device **112** have many of the same features as the article of footwear **10** and device **12**. Many of the components of the article of footwear **110** differ only in scale from corresponding components of the article of footwear **10**, as the article of footwear **10** is sized for an older child, such as a child of grade school age while the article of footwear **110** is sized for a younger child, such as a child of preschool age. Features of the article of footwear **110** and the device **112** that function identically as those described with respect to the article of footwear **10** and device **12** are referred to with identical reference numbers and are as described with respect to the article of footwear **10** and

device **12**. FIG. 25 shows the lateral side arm **30C**, the second joint **37**, the lateral base arm **32C**, and the lateral side protrusion **94**.

[0148] The article of footwear **110** includes a bladder **152** that has a different configuration than the bladder **52**. For example, as shown in FIGS. 28, 37, and 38, the bladder **152** includes a central interior cavity **153** in addition to the arcuate perimeter interior cavity **55**. Webbing **84** surrounds and separates the cavity central interior cavity **153** from the arcuate perimeter interior cavity **55**. The bladder **152** is also configured so that the flange **82** is offset more toward a lower side of the bladder **152** than in the bladder **52**. This causes the portion of the bladder **152** below the flange **82** to be less rounded than the corresponding portion of the bladder **52**. This is evident in FIGS. 25, 39, and 40. As a result, the first outsole component **56** appears relatively flat in FIG. 25 (in comparison to FIG. 1) where it is secured to the lower surface of the bladder **152** as it need not extend as far upward along the bladder **152** to reach the flange **82** given that the flange **82** is offset toward the ground more than in the bladder **52**. The flatter profile of the bladder **152** at the lower side enables both it and the first outsole component **56** to have a wider ground-engaging surface in the transverse direction on a level plane than if the bladder **152** were more rounded at the lower side. This provides greater stability that may be advantageous for the typically younger children wearing the smaller article of footwear **110**.

[0149] Due to the different configuration of the bladder **152**, as best seen in FIGS. 41 and 42, the midsole **154** included in the article of footwear **110** has a bottom surface **80** with a recess **163** shaped to receive not only the inflation port **161** of the bladder **152**, but also the portion of the bladder at the central interior cavity **153**. FIG. 28 shows the recess **163** in the midsole **154** with the bladder **152** able to nest in the recess **163** with the midsole **154** bonded to the upper surface **57** of the bladder **152**.

[0150] FIG. 26 is a medial side view of the article of footwear **110** of FIG. 25, showing the medial side arm **30B**, the first joint **35**, the medial base arm **32B**, and the medial side protrusion **92**. FIG. 27 is a rear view of a portion of the article of footwear **110** of FIG. 25 and with a loaded, access position of the device **112** shown in phantom at **112A**. FIG. 28 is a cross-sectional view of the article of footwear **110** taken at lines 28-28 in FIG. 48 and showing a loaded position of the device **112** in phantom at **112A**.

[0151] FIG. 29 is a cross-sectional view of the article of footwear **110** of FIG. 25 taken at lines 29-29 in FIG. 48. FIG. 30 is a cross-sectional view of the article of footwear **110** taken at lines 30-30 in FIG. 48. FIG. 31 is a cross-sectional view of the article of footwear **110** taken at lines 31-31 in FIG. 48. FIG. 32 is a cross-sectional view of the article of footwear **110** taken at lines 32-32 in FIG. 48. The different shape of the bladder **152** in comparison to the bladder **52** is evident in FIGS. 31 and 32.

[0152] FIG. 33 is a perspective view of the lateral side **50** of the device **112** included in the article of footwear **110**. FIG. 34 is a perspective view of the medial side **48** of the device **112** included in the article of footwear **110**. FIG. 35 is a top view of the device **112**. FIG. 36 is a bottom view of the device **112** showing the rear walls **116**, **120** at the respective concave rear surfaces **114**, **118** at which the convex front surfaces **122**, **124** of the bladder **152** shown in FIG. 37 nest.

[0153] FIG. 37 is a perspective top view and FIG. 38 is a perspective bottom view of the bladder **152** indicating the central interior cavity **153** in addition to the arcuate perimeter interior cavity **55**. FIG. 39 is a perspective view of the medial side of the device **112** secured to a top surface of the bladder **152**, and FIG. 40 is a perspective view of the lateral side of the device **112** secured to the top surface of the bladder **152**. FIGS. 39-40 show the convex front surfaces **122**, **124** of the bladder **152** nested at the respective concave rear surfaces **114**, **118** of the rear walls **116**, **120** of the medial and lateral anchor arms **34**, **36**.

[0154] FIGS. 41-43 indicate the medial recess **102** and the lateral recess **106** at the exterior surface **104** of the midsole **154**, as well as the recess **163** in the bottom surface **80** of the midsole **154**. FIG. 44 shows the bladder **152** to indicate the midsole **154** bonded to the upper surface **57** of the bladder

152. The device **112** is not shown in FIG. **44**. FIGS. **45** and **46** show the midsole **154** with the device **112** having the medial and lateral anchor arms **34**, **36** nested in the recesses **102**, **106**. The bladder **152** is shown in FIG. **47** underlying the base **32** of the device **112** with the medial and lateral anchor arms **34**, **36** extending forward of the bladder **152**.

[0155] In FIG. **48**, the first outsole component **56** is secured to the bottom surface of the bladder **152** and covers the bottom of the arcuate peripheral inflated portion of the bladder **152** shown in FIGS. **37** and **38** so that the bladder **152** is not visible in the bottom view of FIG. **48**. The first outsole component **56** is also secured to a portion of the bottom of the medial and lateral anchor arms **34**, **36**. The second outsole component **58** is secured to the midsole **154** forward of the bladder **152** and inward of the first outsole component **56** and the base **32** shown in FIG. **46**. The second outsole component **58** covers the webbing **84** and the bottom of the bladder **152** at the central interior cavity **153**. As is evident in FIG. **48**, when the first outsole component **56** is secured to the bladder **152** and to the medial anchor arm **34** and lateral anchor arm **36**, a forwardmost portion of the medial anchor arm **34** and a forwardmost portion of the lateral anchor arm **36** are not covered by the first outsole component **56** and are exposed at the ground-facing surface of the sole structure **18**.

[0156] FIG. **49** is a lateral side view of an alternative embodiment of an article of footwear **210** having a device **212** for easing foot entry into the foot-receiving cavity **14** of the article of footwear **210**. The device **212** is shown in an unloaded, use position. The article of footwear **210** and the device **212** have many of the same features as the articles of footwear **10**, **110** and devices **12**, **112**. Many of the components of the article of footwear **210** differ only in scale from corresponding components of the articles of footwear **10**, **110**, as the article of footwear **10** is sized for an older child, such as a child of grade school age while the article of footwear **210** is sized for a toddler. Features of the article of footwear **210** and the device **212** that function identically as those described with respect to the article of footwear **10** and device **12** are referred to with identical reference numbers and are as described with respect to the article of footwear **10** and device **12**. FIG. **49** shows the lateral side arm **30C**, the second joint **37**, the lateral base arm **32C**, and the lateral side projection **94**.

[0157] The article of footwear **210** includes an upper **216** like upper **16** except that the upper **216** includes elastic bands **284** rather than a lace **64**. The elastic bands **284** resiliently stretch to allow foot insertion into the upper **216** and to adjust to the girth of the wearer's foot but do not have ends that require tying as does the lace **64**.

[0158] The article of footwear **210** includes a bladder **252** that has a different configuration than the bladder **52** and the bladder **152**. For example, as shown in FIGS. **52**, **61**, and **62**, the bladder **252** includes a single interior cavity **255** traversed by three areas of webbing **84**. Like bladder **52**, the bladder **252** is also configured so that the flange **82** is offset more toward a lower side of the bladder **252** than in the bladder **52**. This causes the portion of the bladder **252** below the flange **82** to be less rounded than the corresponding portion of the bladder **52**. This is evident in FIGS. **49**, **63**, and **64**. As a result, the first outsole component **56** appears relatively flat in FIG. **49** (in comparison to FIG. **1**) where it is secured to the lower surface of the bladder **252** as it need not extend as far upward along the bladder **252** to reach the flange **82** given that the flange **82** is offset toward the ground more than in the bladder **52**. The flatter profile of the bladder **252** at the lower side enables both it and the first outsole component **56** to have a wider ground-engaging surface in the transverse direction on a level plane than if the bladder **252** were more rounded at the lower side. This provides greater stability that may be advantageous for the typically younger children wearing the smaller article of footwear **210**.

[0159] Due to the different configuration of the bladder **252**, as best seen in FIGS. **65** and **66**, the midsole **254** included in the article of footwear **210** has a bottom surface **80** with a recess **163** shaped to receive the bladder **252**. FIG. **52** shows the recess **263** in the midsole **254** with the bladder **252** able to nest in the recess **263** with the midsole **254** bonded to the upper surface **57** of

the bladder **252**.

[0160] FIG. **50** is a medial side view of the article of footwear **210** of FIG. **49**, showing the medial side arm **30B**, the first joint **35**, the medial base arm **32B**, and the medial side projection **92**. FIG. **51** is a rear view of a portion of the article of footwear **110** of FIG. **49** and with a loaded, access position of the device **212** shown in phantom at **212A**. FIG. **52** is a cross-sectional view of the article of footwear **210** taken at lines **52-52** in FIG. **72** and showing a loaded position of the device **212** in phantom at **212A**.

[0161] FIG. **53** is a cross-sectional view of the article of footwear **210** taken at lines **53-53** in FIG. **72**. FIG. **54** is a cross-sectional view of the article of footwear **110** taken at lines **30-30** in FIG. **48**. FIG. **54** is a cross-sectional view of the article of footwear **210** taken at lines **54-54** in FIG. **72**. FIG. **55** is a cross-sectional view of the article of footwear **210** taken at lines **55-55** in FIG. **72**. The different shape of the bladder **252** in comparison to the bladder **52** is evident in FIGS. **55** and **56**.

[0162] FIG. **57** is a perspective view of the lateral side **50** of the device **212** included in the article of footwear **210**. FIG. **58** is a perspective view of the medial side **48** of the device **212** included in the article of footwear **210**. FIG. **59** is a top view of the device **212**. FIG. **60** is a bottom view of the device **212** showing the rear walls **116**, **120** at the respective concave rear surfaces **114**, **118** at which the convex front surfaces **122**, **124** of the bladder **252** shown in FIG. **61** nest.

[0163] FIG. **61** is a perspective top view and FIG. **62** is a perspective bottom view of the bladder **252** indicating the single interior cavity **255**. FIG. **63** is a perspective view of the medial side of the device **212** secured to a top surface of the bladder **252**, and FIG. **64** is a perspective view of the lateral side of the device **212** secured to the top surface of the bladder **252**. FIGS. **63-64** show the convex front surfaces **122**, **124** of the bladder **252** nested at the respective concave rear surfaces **114**, **118** of the rear walls **116**, **120** of the medial and lateral anchor arms **34**, **36**.

[0164] FIGS. **65-67** indicate the medial recess **102** and the lateral recess **106** at the exterior surface **104** of the midsole **254**, as well as the recess **263** in the bottom surface **80** of the midsole **254**. FIG. **68** shows the bladder **252** to indicate the midsole **254** bonded to the upper surface **57** of the bladder **252**. The device **212** is not shown in FIG. **68**. FIGS. **69** and **70** show the midsole **254** with the device **212** having the medial and lateral anchor arms **34**, **36** nested in the recesses **102**, **106**. The bladder **252** is shown in FIG. **71** underlying the base **32** of the device **212** with the medial and lateral anchor arms **34**, **36** extending forward of the bladder **252**.

[0165] In FIG. **72**, the first outsole component **56** is secured to the bottom surface of the bladder **252** and covers the bottom of the inflated portion of the bladder **252** so that the bladder **252** is not visible in the bottom view of FIG. **72**. The first outsole component **56** is also secured to a portion of the bottom of the medial and lateral anchor arms **34**, **36**. The second outsole component **58** is secured to the midsole **254** forward of the bladder **252** and inward of the first outsole component **56** and the base **32** shown in FIG. **72**. The second outsole component **58** covers the bottom of the bladder **252** inward of the first outsole component **56**. As is evident in FIG. **72**, when the first outsole component **56** is secured to the bladder **252** and to the medial anchor arm **34** and lateral anchor arm **36**, a forwardmost portion of the medial anchor arm **34** and a forwardmost portion of the lateral anchor arm **36** are not covered by the first outsole component **56** and are exposed at the ground-facing surface of the sole structure **18**.

[0166] FIG. **73** is a perspective view of the medial side **48** of the device **12** of the article of footwear of FIG. **1** secured to a foam cushioning component **352** rather than a bladder. FIG. **74** is a perspective view of a lateral side **50** of the device **12** and foam cushioning component **352** of FIG. **73**. The foam cushioning component **352** is shown as a heel cushioning component that has the same volume and shape as the bladder **52**. This enables the foam cushioning component **352** to be incorporated into the article of footwear **10** in place of the bladder **52** and in the same position as the bladder **52** relative to the other footwear components. For example, when incorporated into the article of footwear **10** in place of the bladder **52**, the device **12** is bonded to the top surface of the foam cushioning component **352** with the device **12** and anchoring arms **34**, **36** extending forward

of the foam cushioning component **352**, and the midsole **54** is bonded to the foam cushioning component **352** inward of the base **32**. Alternatively, the foam cushioning component **352** could be configured with a different shape and volume than the bladder **52**. For example, the foam cushioning component **352** need not be configured to have the shape of the bladder **52** at the arcuate interior cavity **55** of the bladder **52** and could instead extend transversely from the medial side to the lateral side without narrowing at a portion corresponding with the webbing **84** of the bladder **52**. Additionally, the foam cushioning component **352** and the midsole **54** could be integrated as a single foam component rather than discrete components like the bladder **52** and the midsole **54**.

[0167] FIG. **75** is an alternative embodiment of an article of footwear **410** that includes many of the same components and features of the article of footwear **10** of FIG. **1**. For example, the article of footwear **410** includes the device **12**, the footwear upper **16**, and the bladder **52** configured and functioning as described with respect to FIG. **1**. As in the article of footwear **10**, the medial anchor arm **34** and the lateral anchor arm **36** of the device **12** are outwardly exposed at the medial side **48** and the lateral side **50** of the article of footwear **410**, respectively.

[0168] The article of footwear **410** has a sole structure **418** that includes the bladder **52** of the sole structure **18**, but differs in some aspects from the sole structure **18**. For example, rather than the single midsole **54**, the sole structure **418** includes an upper midsole layer **454A** (shown in FIG. **78**) and a lower midsole layer **454B** (shown in FIG. **79**). Like the midsole **54**, the upper midsole layer **454A** extends forward of the base **32** and forward of the entire device **12**. The medial anchor arm **34** (shown in FIG. **2**) and the lateral anchor arm **36** extend forward of the bladder **52** and are secured to the upper midsole layer **454A** immediately forward of the bladder **52**. As shown in FIG. **77**, the upper midsole layer **454A** extends rearwardly and transversely inwardly of the base **32** to a rear portion of the base **32** (e.g., to the inner side of the center segment **32A** of the base **32**) and is secured to the upper surface **57** of the bladder **52** forward of the center segment **32A** of the base **32**.

[0169] The lower midsole layer **454B** is disposed inward of the bladder **52** and is secured to a lower surface **76** of the bladder **52** in place of the second outsole component **58** of FIG. **4**. The lower midsole layer **454B** extends forward to the forefoot region **26** and is secured to the bottom surface **480** of the upper midsole layer **454A**. Like the midsole **54**, the upper midsole layer **454A** includes the medial recess **102** in an exterior surface **404** of the upper midsole layer **454A** and the lateral recess **106** in the exterior surface **404** and the device **12** is configured to fit to the upper midsole layer **454A** within the recesses **102**, **106**, with the medial anchor arm **34** nesting in the medial recess **102** and the lateral anchor arm **36** nesting in the lateral recess **106**. Accordingly, the device **12** is thus secured to the upper **16**, to the bladder **52**, and to the upper midsole layer **454A** and wraps around the upper **16** from the rear of the upper **16**, along the medial and lateral sides **48**, **50**, and extends forward of the bladder **52**.

[0170] A first outsole component **456** is secured to the bottom surface of the bladder **52** and to bottom surfaces of the medial anchor arm **34** and lateral anchor arm **36**. The first outsole component **456** differs from the first outsole component **56** in that it extends forward of the device **12** and is secured to the bottom surface **480** of the upper midsole layer **454A** in the forefoot region **26**, so that the forwardmost portion of the medial anchor arm **34** and the forwardmost portion of the lateral anchor arm **36** are covered by the first outsole component **456** rather than being exposed at the ground-facing surface as in the sole structure **18**. For example, a medial terminal end **456A** and a lateral terminal end **456B** of the first outsole component **456** are both disposed in the forefoot region **26**. The first outsole component **456** thus protects the anchor arms **34**, **36**. Additionally, because the medial terminal end **456A** and the lateral terminal end **456B** are in the forefoot region **26** under the metatarsal heads of the foot, flexibility during dorsiflexion is not inhibited by the first outsole component **456**. Stated differently, the first outsole component **456** is decoupled from a third outsole component **460**, discussed below, at an expected bending axis during dorsiflexion.

[0171] A second outsole component **458** is secured to a bottom surface **481** of the lower midsole

layer **454B** in the midfoot region **24** and forefoot region **26** forward of the bladder **52**. The second outsole component **458** does not extend to the heel region **22**, so that the lower midsole layer **454B** is exposed at the ground-facing surface of the sole structure **418** rearward of the second outsole component **458** and inward of a first outsole component **456**.

[0172] A third outsole component **460** is disposed in the forefoot region **26** forward of the terminal ends **456A**, **456B** of the first outsole component **456** and forward of the second outsole component **458**. A portion of the lower midsole layer **454B** is exposed between the second outsole component **458** and the third outsole component **460** at the ground-facing surface. In fact, an entire outer peripheral edge of the second outsole component **458** is decoupled from and spaced apart from the first outsole component **456** and the third outsole component **460**. The decoupling of the second outsole component **458** from both the first outsole component **456** and the third outsole component **460** may allow for greater resilient deformation of the stacked midsole layers **454A**, **454B** in the forefoot region **22**.

[0173] FIG. **81** is an alternative embodiment of an article of footwear **510** that includes many of the same components and features of the article of footwear **110** of FIG. **25**. For example, the article of footwear **510** includes the device **112**, the footwear upper **16**, and the bladder **152** configured and functioning as described with respect to FIG. **25**. As in the article of footwear **110**, the medial anchor arm **34** and the lateral anchor arm **36** of the device **112** are outwardly exposed at the medial side **48** and the lateral side **50** of the article of footwear **510**, respectively.

[0174] The article of footwear **510** has a sole structure **518** that includes the bladder **52** of the sole structure **18**, but differs in some aspects from the sole structure **18**. For example, rather than the single midsole **154**, the sole structure **518** includes an upper midsole layer **554A** (shown in FIG. **84**) and a lower midsole layer **554B** (shown in FIG. **85**). Like the midsole **154**, the upper midsole layer **554A** extends forward of the base **32** and forward of the entire device **112**. The medial anchor arm **34** (shown in FIG. **26**) and the lateral anchor arm **36** extend forward of the bladder **152** and are secured to the upper midsole layer **554A** immediately forward of the bladder **52**. As shown in FIG. **83**, the upper midsole layer **554A** extends rearwardly and transversely inwardly of the base **32** to a rear portion of the base **32** (e.g., to the inner side of the center segment **32A** of the base **32**) and is secured to the upper surface **57** of the bladder **152** forward of the center segment **32A** of the base **32**.

[0175] The lower midsole layer **554B** is disposed inward of the bladder **152**. The lower midsole layer **554B** extends forward to the forefoot region **26** and is secured to the bottom surface **580** of the upper midsole layer **554A**. Like the midsole **154**, the upper midsole layer **554A** includes the medial recess **102** in an exterior surface **504** of the upper midsole layer **554A** and the lateral recess **106** in the exterior surface **504** and the device **112** is configured to fit to the upper midsole layer **554A** within the recesses **102**, **106**, with the medial anchor arm **34** nesting in the medial recess **102** and the lateral anchor arm **36** nesting in the lateral recess **106**. Accordingly, the device **112** is thus secured to the upper **16**, to the bladder **152**, and to the upper midsole layer **554A** and wraps around the upper **16** from the rear of the upper **16**, along the medial and lateral sides **48**, **50**, and extends forward of the bladder **152**.

[0176] A first outsole component **556** is secured to the bottom surface of the bladder **152** and to bottom surfaces of the medial anchor arm **34** and lateral anchor arm **36**. The first outsole component **556** differs from the first outsole component **56** in that it extends forward of the device **12** and is secured to the bottom surface **580** of the upper midsole layer **554A** in the forefoot region **26**, so that the forwardmost portion of the medial anchor arm **34** and the forwardmost portion of the lateral anchor arm **36** are covered by the first outsole component **556** rather than being exposed at the ground-facing surface as in the sole structure **18**. For example, a medial terminal end **556A** and a lateral terminal end **556B** of the first outsole component **556** are both disposed in the forefoot region **26**. The first outsole component **556** thus protects the anchor arms **34**, **36**. Additionally, because the medial terminal end **556A** and the lateral terminal end **556B** are in the forefoot region

26 under the metatarsal heads of the foot, flexibility during dorsiflexion is not inhibited by the first outsole component **556**. Stated differently, the first outsole component **556** is decoupled from a third outsole component **560**, discussed below, at an expected bending axis during dorsiflexion. [0177] A second outsole component **558** is secured to a bottom surface **581** of the lower midsole layer **554B** in the midfoot region **24** and forefoot region **26** forward of the bladder **152**. The second outsole component **558** does not extend to the heel region **22**, so that the lower midsole layer **554B** is exposed at the ground-facing surface of the sole structure **518** rearward of the second outsole component **558** and inward of the first outsole component **556**.

[0178] The lower midsole layer **554B** extends under and is secured to a central portion **556C** of the first outsole component **556**. FIG. **87** shows that the top side of the central portion **556C** is contoured to fit to and be secured to the lower surface **76** of the bladder **152**.

[0179] A third outsole component **560** is disposed in the forefoot region **26** forward of the terminal ends **556A**, **556B** of the first outsole component **556** and forward of the second outsole component **558**. A portion of the lower midsole layer **554B** is exposed between the second outsole component **558** and the third outsole component **560** at the ground-facing surface. In fact, an entire outer peripheral edge of the second outsole component **558** is decoupled from and spaced apart from the first outsole component **556** and the third outsole component **560**. The decoupling of the second outsole component **558** from both the first outsole component **556** and the third outsole component **560** may allow for greater resilient deformation of the stacked midsole layers **554A**, **554B** in the forefoot region **22**.

[0180] FIG. **88** is an alternative embodiment of an article of footwear **610** that includes many of the same components and features of the article of footwear **210** of FIG. **49**. For example, the article of footwear **610** includes the device **212**, the footwear upper **216**, and the bladder **252** configured and functioning as described with respect to FIG. **49**. As in the article of footwear **210**, the medial anchor arm **34** and the lateral anchor arm **36** of the device **212** are outwardly exposed at the medial side **48** and the lateral side **50** of the article of footwear **610**, respectively.

[0181] The article of footwear **610** has a sole structure **618** that includes the bladder **652** of the sole structure **18** of the article of footwear **210**, but differs in some aspects from the sole structure **18**. For example, a single midsole **654** shown in FIG. **91** is similar to the midsole **254**. Like the midsole **254**, the midsole **654** extends forward of the base **32** and forward of the entire device **212**. The medial anchor arm **34** (shown in FIG. **50**) and the lateral anchor arm **36** extend forward of the bladder **252** and are secured to the midsole **654** immediately forward of the bladder **252**. As shown in FIG. **90**, the midsole **654** extends rearwardly and transversely inwardly of the base **32** to a rear portion of the base **32** (e.g., to the inner side of the center segment **32A** of the base **32**) and is secured to the upper surface **57** of the bladder **252** forward of the center segment **32A** of the base **32**.

[0182] The midsole **654** is disposed inward of the bladder **252** and extends forward to the forefoot region **26**. Like the midsole **254**, the midsole **654** includes the medial recess **102** and the lateral recess **106** in an exterior surface **604** of the midsole layer **654** and the device **212** is configured to fit to the midsole **654** within the recesses **102**, **106**, with the medial anchor arm **34** nesting in the medial recess **102** and the lateral anchor arm **36** nesting in the lateral recess **106**. Accordingly, the device **212** is thus secured to the upper **216**, to the bladder **252**, and to midsole **654** and wraps around the upper **216** from the rear of the upper **216**, along the medial and lateral sides **48**, **50**, and extends forward of the bladder **252**.

[0183] A first outsole component **656** is secured to the bottom surface of the bladder **652** and to bottom surfaces of the medial anchor arm **34** and lateral anchor arm **36**. The first outsole component **656** differs from the first outsole component **56** in that it extends forward of the device **212** and is secured to the bottom surface **680** of the midsole **654** in the forefoot region **26**, so that the forwardmost portion of the medial anchor arm **34** and the forwardmost portion of the lateral anchor arm **36** are covered by the first outsole component **656** rather than being exposed at the

ground-facing surface as in the sole structure **18**. For example, a medial terminal end **656A** and a lateral terminal end **656B** of the first outsole component **656** are both disposed in the forefoot region **26**. The first outsole component **656** thus protects the anchor arms **34**, **36**. Additionally, because the medial terminal end **656A** and the lateral terminal end **656B** are in the forefoot region **26** under the metatarsal heads of the foot, flexibility during dorsiflexion is not inhibited by the first outsole component **656**. Stated differently, the first outsole component **656** is decoupled from a third outsole component **660**, discussed below, at an expected bending axis during dorsiflexion.

[0184] A second outsole component **658** is secured to a bottom surface **680** of the midsole **654** in the midfoot region **24** and forefoot region **26** forward of the bladder **252**. The second outsole component **658** extends to the heel region **22** where it is secured to a lower side of the central portion **656C** of the first outsole component **656**. Stated differently, the second outsole component **658** extends under and is secured to the central portion **656C** of the first outsole component **656**.

[0185] A third outsole component **660** is disposed in the forefoot region **26** forward of the terminal ends **656A**, **656B** of the first outsole component **656** and forward of the second outsole component **658**. A portion of the midsole **654** is exposed between the second outsole component **658** and the third outsole component **660** at the ground-facing surface. In fact, an entire outer peripheral edge of the second outsole component **658** is decoupled from and spaced apart from the first outsole component **656** and the third outsole component **660**. The decoupling of the second outsole component **658** from both the first outsole component **656** and the third outsole component **660** may allow for greater resilient deformation of the **6** midsole **654** in the forefoot region **22**.

[0186] The following Clauses provide example configurations of a device for easing foot entry into an article of footwear and of an article of footwear disclosed herein.

[0187] Clause 1. A device for easing foot entry into an article of footwear and configured to surround a portion of a foot-receiving void at a heel region of an article of footwear, the device comprising: a control bar having a center segment, a medial side arm extending downwardly and forwardly from the center segment, and a lateral side arm spaced from the medial side arm and extending downwardly and forwardly from the center segment; a base underlying the control bar and connected to the medial side arm at a first joint and to the lateral side arm at a second joint; wherein the control bar and the base are configured to surround a rear of the heel region with the medial side arm at a medial side of the article of footwear and the lateral side arm at a lateral side of the article of footwear; a medial anchor arm extending downwardly and forwardly from the first joint and partially under the foot-receiving void; and a lateral anchor arm extending downwardly and forwardly from the second joint and partially under the foot-receiving void; wherein the medial side arm and the lateral side arm resiliently bend to a loaded position when a downward force is applied to the center segment of the control bar, storing potential energy that returns the control bar to an unloaded position upon removal of the downward force.

[0188] Clause 2. The device of clause 1, wherein the base has a medial side projection extending transversely outwardly from a medial side of the base rearward of the first joint.

[0189] Clause 3. The device of any of clauses 1-2, wherein the base has a lateral side projection extending transversely outwardly from a lateral side of the base rearward of the second joint.

[0190] Clause 4. The device of any of clauses 1-3, wherein the base has a rear projection extending rearwardly from a rear of the base.

[0191] Clause 5. The device of clause 4, wherein an exterior rear surface of the base above the rear projection defines a concavity.

[0192] Clause 6. The device of any of clauses 1-5, wherein the device includes a ledge disposed at an upper extent of the center segment of the control bar.

[0193] Clause 7. The device of any of clauses 1-6, wherein an exterior rear surface of the base defines a concavity.

[0194] Clause 8. The device of any of clauses 1-7, wherein a lower surface of the base has an inner edge and an outer edge and is concave in a transverse direction of the article of footwear and in a

longitudinal direction of the article of footwear from the inner edge to the outer edge, with the outer edge lower than the inner edge.

[0195] Clause 9. The device of any of clauses 1-8, wherein the device is a single, unitary, one-piece component.

[0196] Clause 10. An article of footwear including the device of any of clauses 1-9.

[0197] Clause 11. An article of footwear comprising: an upper; a sole structure secured to and underlying the upper, the upper and the sole structure defining a foot-receiving void; a device including: a control bar having a center segment, a medial side arm extending downwardly and forwardly from the center segment, and a lateral side arm spaced from the medial side arm and extending downwardly and forwardly from the center segment; a base underlying the control bar and connected to the medial side arm at a first joint and to the lateral side arm at a second joint; wherein the control bar and the base are configured to surround a rear of the upper with the medial side arm at a medial side of the upper, the lateral side arm at a lateral side of the upper and the base secured to an upper surface of the sole structure; a medial anchor arm extending downwardly and forwardly from the first joint and partially under the sole structure and the foot-receiving void; and a lateral anchor arm extending downwardly and forwardly from the second joint and partially under the sole structure and the foot-receiving void; wherein the medial side arm and the lateral side arm resiliently bend to a loaded position when a downward force is applied to the center segment of the control bar, storing potential energy that returns the control bar to an unloaded position upon removal of the downward force, the rear of the upper moving with the center segment of the control bar.

[0198] Clause 12. The article of footwear of clause 11, wherein the sole structure includes a bladder defining an interior cavity and configured to retain a fluid in the interior cavity; and wherein the base is disposed on the bladder.

[0199] Clause 13. The article of footwear of clause 12, wherein the base is secured to an upper surface of the bladder.

[0200] Clause 14. The article of footwear of clause 13, wherein the upper surface of the bladder is a convex upper surface and a lower surface of the base is concave and interfits with and is secured to the convex upper surface of the bladder.

[0201] Clause 15. The article of footwear of any of clauses 13-14, wherein the

[0202] base has a medial side projection extending transversely outwardly and downwardly over the bladder from a medial side of the base rearward of the first joint; and the medial side projection is not fixed to the bladder.

[0203] Clause 16. The article of footwear of any of clauses 13-15, wherein the base has a lateral side projection extending transversely outwardly and downwardly over the bladder from a lateral side of the base rearward of the second joint; and the lateral side projection is not fixed to the bladder.

[0204] Clause 17. The article of footwear of any of clauses 13-16, wherein the

[0205] base has a rear projection extending rearwardly and downwardly over the bladder from a rear of the base; and the rear projection is not fixed to the bladder.

[0206] Clause 18. The article of footwear of clause 17, wherein an exterior rear surface of the base above the rear projection defines a concavity.

[0207] Clause 19. The article of footwear of any of clauses 13-18, wherein: the sole structure includes a midsole extending forward of the base; the medial anchor arm nests in a medial recess in an exterior surface of the midsole and is outwardly exposed; and the lateral anchor arm nests in a lateral recess in the exterior surface of the midsole and is outwardly exposed.

[0208] Clause 20. The article of footwear of clause 19, wherein the midsole extends rearwardly and transversely inwardly of the base to a rear portion of the base and is secured to the upper surface of the bladder.

[0209] Clause 21. The article of footwear of any of clauses 13-20, wherein: a rear surface of the

medial anchor arm defines a downwardly extending rear wall and a front surface of the bladder nests against the rear wall of the medial anchor arm; and a rear surface of the lateral anchor arm defines a downwardly extending rear wall and the front surface of the bladder nests against the rear wall of the lateral anchor arm.

[0210] Clause 22. The article of footwear of clause 21, wherein the sole structure further includes an outsole component secured to a bottom surface of the bladder rearward of the medial anchor arm and the lateral anchor arm.

[0211] Clause 23. The article of footwear of clause 22, wherein the outsole component is further secured to a bottom surface of the medial anchor arm and a bottom surface of the lateral anchor arm forward of the bladder.

[0212] Clause 24. The article of footwear of clause 22, wherein: the sole structure includes a midsole secured to an upper surface of the bladder and extending forward of the base; and the outsole component is a first outsole component, the bottom surface of the bladder and the first outsole component are arcuate, and the sole structure further includes a second outsole component secured to the bladder inward of the first outsole component and extending forward under the midsole and secured to a bottom surface of the midsole forward of the bladder.

[0213] Clause 25. The article of footwear of any of clauses 11-24, wherein a portion of the medial anchor arm and a portion of the lateral anchor arm are exposed at a ground-facing surface of the sole structure.

[0214] Clause 26. The article of footwear of any of clauses 11-25, wherein an exterior rear surface of the base defines a concavity.

[0215] Clause 27. The article of footwear of any of clauses 11-26, wherein: the sole structure includes a midsole extending forward of the base; the medial anchor arm nests in a medial recess in an exterior surface of the midsole and is outwardly exposed; and the lateral anchor arm nests in a lateral recess in the exterior surface of the midsole and is outwardly exposed.

[0216] Clause 28. The article of footwear of clause 27, wherein the midsole extends rearwardly and transversely inwardly of the base to a rear portion of the base.

[0217] Clause 29. The article of footwear of any of clauses 11-28, wherein the device includes a ledge disposed at an upper extent of the center segment of the control bar.

[0218] Clause 30. The article of footwear of clause 29, wherein the upper includes a cushioned collar component above the center segment of the control bar and the ledge.

[0219] Clause 31. The article of footwear of clause 30, wherein the cushioned collar component comprises foam.

[0220] Clause 32. The article of footwear of any of clauses 11-31, wherein the upper includes a single layer between the center segment of the control bar and the base.

[0221] Clause 33. The article of footwear of any of clauses 11-32, wherein the

[0222] device defines a gap between the control bar and the base extending around the rear of the upper from the first joint to the second joint and the article of footwear is characterized by an absence of a heel counter.

[0223] Clause 34. The article of footwear of any of clauses 11-33, wherein the center segment of the control bar has a thinned portion; and the upper is stitched to the control bar at the thinned portion.

[0224] Clause 35. The article of footwear of clause 34, wherein the medial side arm and the lateral side arm are not fixed to the upper.

[0225] Clause 36. The article of footwear of any of clauses 11-35, wherein the upper includes a tongue portion, and the article of footwear further comprising: at least one elastically stretchable band secured to the upper and extending over the tongue portion.

[0226] Clause 37. The article of footwear of any of clauses 11-36, wherein the device is a single, unitary, one-piece component.

[0227] Clause 38. An article of footwear comprising: an upper; a sole structure secured to and

underlying the upper, the upper and the sole structure defining a foot-receiving void; and a device including: a control bar having a center segment, a medial side arm extending downwardly and forwardly from the center segment, and a lateral side arm spaced from the medial side arm and extending downwardly and forwardly from the center segment; a base underlying the control bar and connected to the medial side arm at a first joint and to the lateral side arm at a second joint; wherein the control bar and the base are configured to surround a rear of the upper with the medial side arm at a medial side of the upper, the lateral side arm at a lateral side of the upper; wherein the sole structure includes a bladder defining an interior cavity and configured to retain a fluid in the interior cavity; wherein the base is secured to an upper surface of the bladder; wherein the medial side arm and the lateral side arm resiliently bend to a loaded position when a downward force is applied to the center segment of the control bar, storing potential energy that returns the control bar to an unloaded position upon removal of the downward force, the rear of the upper moving with the center segment of the control bar.

[0228] Clause 39. The article of footwear of clause 38, wherein the upper surface of the bladder is a convex upper surface and a lower surface of the base is concave and interfits with and is secured to the convex upper surface of the bladder.

[0229] Clause 40. The article of footwear of any of clauses 38-39, wherein the base has a medial side projection extending transversely outwardly and downwardly over the bladder from a medial side of the base rearward of the first joint; and the medial side projection is not fixed to the bladder.

[0230] Clause 41. The article of footwear of any of clauses 38-40, wherein the base has a lateral side projection extending transversely outwardly and downwardly over the bladder from a lateral side of the base rearward of the second joint; and the lateral side projection is not fixed to the bladder.

[0231] Clause 42. The article of footwear of any of clauses 38-41, wherein the base has a rear projection extending rearwardly and downwardly over the bladder from a rear of the base; and the rear projection is not fixed to the bladder.

[0232] Clause 43. The article of footwear of clause 42, wherein an exterior rear surface of the base above the rear projection defines a concavity.

[0233] Clause 44. The article of footwear of any of clauses 38-43, the device further including: a medial anchor arm extending downwardly and forwardly from the first joint and partially under the sole structure and the foot-receiving void; and a lateral anchor arm extending downwardly and forwardly from the second joint and partially under the sole structure and the foot-receiving void; wherein: the sole structure includes a midsole extending forward of the base; the medial anchor arm nests in a medial recess in an exterior surface of the midsole and is outwardly exposed; and the lateral anchor arm nests in a lateral recess in the exterior surface of the midsole and is outwardly exposed.

[0234] Clause 45. The article of footwear of clause 44, wherein the midsole extends rearwardly and transversely inwardly of the base to a rear portion of the base and is secured to the upper surface of the bladder.

[0235] Clause 46. The article of footwear of any of clauses 38-45, wherein: a rear surface of the medial anchor arm defines a downwardly extending rear wall and a front surface of the bladder nests against the rear wall of the medial anchor arm; and a rear surface of the lateral anchor arm defines a downwardly extending rear wall and the front surface of the bladder nests against the rear wall of the lateral anchor arm.

[0236] Clause 47. The article of footwear of clause 46, wherein the sole structure further includes an outsole component secured to a bottom surface of the bladder rearward of the medial anchor arm and the lateral anchor arm.

[0237] Clause 48. The article of footwear of clause 47, wherein the outsole component is further secured to a bottom surface of the medial anchor arm and a bottom surface of the lateral anchor arm forward of the bladder.

[0238] Clause 49. The article of footwear of any of clauses 47-48, wherein the sole structure includes a midsole secured to an upper surface of the bladder and extending forward of the base; and the outsole component is a first outsole component, the bottom surface of the bladder and the first outsole component are arcuate, and the sole structure further includes a second outsole component secured to the bladder inward of the first outsole component and extending forward under the midsole and secured to a bottom surface of the midsole forward of the bladder.

[0239] Clause 50. The article of footwear of any of clauses 38-49, wherein a portion of the medial anchor arm and a portion of the lateral anchor arm are exposed at a ground-facing surface of the sole structure.

[0240] Clause 51. The article of footwear of any of clauses 38-50, wherein an exterior rear surface of the base defines a concavity.

[0241] Clause 52. The article of footwear of any of clauses 38-51, wherein: the sole structure includes a midsole extending forward of the base; the medial anchor arm nests in a medial recess in an exterior surface of the midsole and is outwardly exposed; and the lateral anchor arm nests in a lateral recess in the exterior surface of the midsole and is outwardly exposed.

[0242] Clause 53. The article of footwear of any of clauses 38-52, wherein the device includes a ledge disposed at an upper extent of the center segment of the control bar.

[0243] Clause 54. The article of footwear of clause 53, wherein the upper includes a cushioned collar component above the center segment of the control bar and the ledge.

[0244] Clause 55. The article of footwear of clause 54, wherein the cushioned collar component comprises foam.

[0245] Clause 56. The article of footwear of any of clauses 38-55, wherein the upper includes a single layer between the center segment of the control bar and the base.

[0246] Clause 57. The article of footwear of any of clauses 38-56, wherein the device defines a gap between the control bar and the base extending around the rear of the upper from the first joint to the second joint and the article of footwear is characterized by an absence of a heel counter.

[0247] Clause 58. The article of footwear of any of clauses 38-57, wherein the center segment of the control bar has a thinned portion; and the upper is stitched to the control bar at the thinned portion.

[0248] Clause 59. The article of footwear of clause 58, the device further including: a medial anchor arm extending downwardly and forwardly from the first joint and partially under the sole structure and the foot-receiving void; and a lateral anchor arm extending downwardly and forwardly from the second joint and partially under the sole structure and the foot-receiving void; wherein the medial side arm and the lateral side arm are not fixed to the upper.

[0249] Clause 60. The article of footwear of any of clauses 38-59, wherein the upper includes a tongue portion, and the article of footwear further comprising: at least one elastically stretchable band secured to the upper and extending over the tongue portion.

[0250] Clause 61. The article of footwear of any of clauses 38-60, wherein the device is a single, unitary, one-piece component.

[0251] To assist and clarify the description of various embodiments, various terms are defined herein. Unless otherwise indicated, the following definitions apply throughout this specification (including the claims). Additionally, all references referred to are incorporated herein in their entirety.

[0252] An “article of footwear”, a “footwear article of manufacture”, and “footwear” may be considered to be both a machine and a manufacture. Assembled, ready to wear footwear articles (e.g., shoes, sandals, boots, etc.), as well as discrete components of footwear articles (such as a midsole, an outsole, an upper component, etc.) prior to final assembly into ready to wear footwear articles, are considered and alternatively referred to herein in either the singular or plural as “article(s) of footwear”.

[0253] “A”, “an”, “the”, “at least one”, and “one or more” are used interchangeably to indicate that

at least one of the items is present. A plurality of such items may be present unless the context clearly indicates otherwise. All numerical values of parameters (e.g., of quantities or conditions) in this specification, unless otherwise indicated expressly or clearly in view of the context, including the appended claims, are to be understood as being modified in all instances by the term “about” whether or not “about” actually appears before the numerical value. “About” indicates that the stated numerical value allows some slight imprecision (with some approach to exactness in the value; approximately or reasonably close to the value; nearly). If the imprecision provided by “about” is not otherwise understood in the art with this ordinary meaning, then “about” as used herein indicates at least variations that may arise from ordinary methods of measuring and using such parameters. In addition, a disclosure of a range is to be understood as specifically disclosing all values and further divided ranges within the range.

[0254] The terms “comprising”, “including”, and “having” are inclusive and therefore specify the presence of stated features, steps, operations, elements, or components, but do not preclude the presence or addition of one or more other features, steps, operations, elements, or components. Orders of steps, processes, and operations may be altered when possible, and additional or alternative steps may be employed. As used in this specification, the term “or” includes any one and all combinations of the associated listed items. The term “any of” is understood to include any possible combination of referenced items, including “any one of” the referenced items. The term “any of” is understood to include any possible combination of referenced claims of the appended claims, including “any one of” the referenced claims.

[0255] For consistency and convenience, directional adjectives may be employed throughout this detailed description corresponding to the illustrated embodiments. Those having ordinary skill in the art will recognize that terms such as “above”, “below”, “upward”, “downward”, “top”, “bottom”, etc., may be used descriptively relative to the figures, without representing limitations on the scope of the invention, as defined by the claims.

[0256] The term “longitudinal” particularly refers to a direction extending a length of a component. For example, a longitudinal direction of a shoe extends between a forefoot region and a heel region of the shoe. The term “forward” or “anterior” is used to particularly refer to the general direction from a heel region toward a forefoot region, and the term “rearward” or “posterior” is used to particularly refer to the opposite direction, i.e., the direction from the forefoot region toward the heel region. In some cases, a component may be identified with a longitudinal axis as well as a forward and rearward longitudinal direction along that axis. The longitudinal direction or axis may also be referred to as an anterior-posterior direction or axis.

[0257] The term “transverse” particularly refers to a direction extending a width of a component. For example, a transverse direction of a shoe extends between a lateral side and a medial side of the shoe. The transverse direction or axis may also be referred to as a lateral direction or axis or a mediolateral direction or axis.

[0258] The term “vertical” particularly refers to a direction generally perpendicular to both the lateral and longitudinal directions. For example, in cases where a sole is planted flat on a ground surface, the vertical direction may extend from the ground surface upward. It will be understood that each of these directional adjectives may be applied to individual components of a sole. The term “upward” or “upwards” particularly refers to the vertical direction pointing towards a top of the component, which may include an instep, a fastening region and/or a throat of an upper. The term “downward” or “downwards” particularly refers to the vertical direction pointing opposite the upwards direction, toward the bottom of a component and may generally point towards the bottom of a sole structure of an article of footwear.

[0259] The “interior” of an article of footwear, such as a shoe, particularly refers to portions at the space that is occupied by a wearer's foot when the shoe is worn. The “inner side” of a component particularly refers to the side or surface of the component that is (or will be) oriented toward the interior of the component or article of footwear in an assembled article of footwear. The “outer

side” or “exterior” of a component particularly refers to the side or surface of the component that is (or will be) oriented away from the interior of the shoe in an assembled shoe. In some cases, other components may be between the inner side of a component and the interior in the assembled article of footwear. Similarly, other components may be between an outer side of a component and the space external to the assembled article of footwear. Further, the terms “inward” and “inwardly” particularly refer to the direction toward the interior of the component or article of footwear, such as a shoe, and the terms “outward” and “outwardly” particularly refer to the direction toward the exterior of the component or article of footwear, such as the shoe. In addition, the term “proximal” particularly refers to a direction that is nearer a center of a footwear component, or is closer toward a foot when the foot is inserted in the article of footwear as it is worn by a user. Likewise, the term “distal” particularly refers to a relative position that is further away from a center of the footwear component or is further from a foot when the foot is inserted in the article of footwear as it is worn by a user. Thus, the terms proximal and distal may be understood to provide generally opposing terms to describe relative spatial positions.

[0260] While various embodiments have been described, the description is intended to be exemplary, rather than limiting and it will be apparent to those of ordinary skill in the art that many more embodiments and implementations are possible that are within the scope of the embodiments. Any feature of any embodiment may be used in combination with or substituted for any other feature or element in any other embodiment unless specifically restricted. Accordingly, the embodiments are not to be restricted except in light of the attached claims and their equivalents. Also, various modifications and changes may be made within the scope of the attached claims.

[0261] While several modes for carrying out the many aspects of the present teachings have been described in detail, those familiar with the art to which these teachings relate will recognize various alternative aspects for practicing the present teachings that are within the scope of the appended claims. It is intended that all matter contained in the above description or shown in the accompanying drawings shall be interpreted as illustrative and exemplary of the entire range of alternative embodiments that an ordinarily skilled artisan would recognize as implied by, structurally and/or functionally equivalent to, or otherwise rendered obvious based upon the included content, and not as limited solely to those explicitly depicted and/or described embodiments.

Claims

1. An article of footwear comprising: an upper; a sole structure secured to and underlying the upper, wherein the sole structure includes a bladder defining an interior cavity and configured to retain a fluid in the interior cavity; and a device including: a control bar having a center segment, a medial side arm extending downwardly and forwardly from the center segment, and a lateral side arm spaced from the medial side arm and extending downwardly and forwardly from the center segment; and a base underlying the control bar and connected to the medial side arm at a first joint and to the lateral side arm at a second joint; wherein the control bar and the base are configured to surround a rear of the upper with the medial side arm at a medial side of the upper, the lateral side arm at a lateral side of the upper; wherein the bladder has a convex upper surface, and the base has a concave lower surface that interfits with and is secured to the convex upper surface of the bladder.
2. The article of footwear of claim 1, wherein the concave lower surface of the base has an inner edge and an outer edge and is concave in a transverse direction of the device and in a longitudinal direction of the device from the inner edge to the outer edge, with the outer edge lower than the inner edge.
3. The article of footwear of claim 1, wherein: the center segment of the control bar has a thinned portion; and the upper is stitched to the control bar at the thinned portion.

4. The article of footwear of claim 1, wherein the medial side arm and the lateral side arm are not fixed to the upper.

5. The article of footwear of claim 1, wherein the medial side arm and the lateral side arm are adapted to resiliently bend to a loaded position when a downward force is applied to the center segment of the control bar, storing potential energy that returns the control bar to an unloaded position upon removal of the downward force.

6. The article of footwear of claim 1, wherein: the upper and the sole structure define a foot-receiving void; and the device includes a medial anchor arm extending downwardly and forwardly from the first joint and adapted to be positioned partially under the foot-receiving void.

7. The article of footwear of claim 6, wherein: the bladder has a convex front surface at a medial side of the bladder; and the convex front surface at the medial side of the bladder is nested in a concave rear surface of the medial anchor arm.

8. The article of footwear of claim 1, wherein: the upper and the sole structure define a foot-receiving void; and the device includes a lateral anchor arm extending downwardly and forwardly from the second joint and adapted to be positioned partially under the foot-receiving void.

9. The article of footwear of claim 8, wherein: the bladder has a convex front surface at a lateral side of the bladder; and the convex front surface at the lateral side of the bladder is nested in a concave rear surface of the lateral anchor arm.

10. The article of footwear of claim 1, wherein the sole structure includes: a midsole secured to an upper surface of the bladder and extending forward of the base; a first outsole component, secured to a bottom surface of the bladder; and a second outsole component secured to the bladder inward of the first outsole component and secured to a bottom surface of the midsole forward of the bladder.

11. The article of footwear of claim 10, wherein an inflated portion of the bladder and the first outsole component are arcuate.

12. The article of footwear of claim 1, wherein: the device includes a medial anchor arm extending downwardly and forwardly from the first joint; the device includes a lateral anchor arm extending downwardly and forwardly from the second joint; and the sole structure further includes an outsole component secured to a bottom surface of the bladder rearward of the medial anchor arm and the lateral anchor arm.

13. The article of footwear of claim 12, wherein the outsole component is further secured to a bottom surface of the medial anchor arm and a bottom surface of the lateral anchor arm forward of the bladder.

14. The article of footwear of claim 13, wherein the outsole component is a first outsole component, and the sole structure further includes: a midsole secured to an upper surface of the bladder and extending forward of the base; and a second outsole component secured to the bladder inward of the first outsole component and secured to a bottom surface of the midsole forward of the bladder.

15. The article of footwear of claim 14, wherein the second outsole component is secured to an uninflated portion of the bladder.

16. The article of footwear of claim 1, wherein the sole structure further includes a midsole secured to an upper surface of the bladder and extending rearwardly and transversely inward of the base to a rear portion of the base.

17. The article of footwear of claim 16, wherein the midsole further extends forward of the base.

18. The article of footwear of claim 1, wherein the device is a single, unitary, one-piece component.

19. The article of footwear of claim 1, wherein the base has: a medial side projection extending transversely outwardly from a medial side of the base rearward of the first joint; and/or a lateral side projection extending transversely outwardly from a lateral side of the base rearward of the second joint; and/or a rear projection extending rearwardly from a rear of the base.

20. The article of footwear of claim 1, wherein the sole structure is characterized by an absence of a heel counter between the center segment and the base.
